AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 34 · STUDENT LESSON

Capstone Dataset and Baseline

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Choose one source. Audit the groups. Run the baseline.

Begin a complete,reproducible capstone with explicit dataset identity and protected final references.

Keep one habit: Actual72-row kit → data dictionary → runnable baseline → grouped plant-plan contrast → Gate1audit → independent related-pair repair.

MAIN LESSON · SECTION 1

Can another learner run your baseline from the actual data?

> Begin the capstone with a bounded question,source-aware split and complete evidence kit.

Learning goals

I can define a feasible capstone question and acceptance criteria.

I can create a data and split plan before modelling.

I can specify **evidence**, privacy and a fallback.

Recall evaluation choices before selection

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Model Cards and Human Review. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Subgroup Performance. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.
- Which evaluation choices should precede model selection?

Readiness for the baseline/candidate extension

- What must be fixed before the capstone comparison?

> **My readiness explanation**
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End goal

By the end, I can explain capstone dataset and **baseline** using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

A responsible capstone begins by freezing the problem, intended use, dataset documentation, grouped split, and **baseline evidence**.



LEARNING TARGET A capstone has two views of each of 40 leaves. Specify a 24/8/8 leaf split and **baseline** procedure. When may final labels be consulted?

Builder name and experiment date:

Readiness for the connected explanation

Readiness for Capstone models and baselines

This week's end goal

MAIN LESSON · SECTION 2

Choose one documented task and build its simple reference first

> The provided kit makes the project runnable without personal records or paid tools.

Capstone scope and dataset-choice explanation

The capstone is a classroom investigation, not a product launch. Choose the supplied fictional clear/blocked parcel task or an equally bounded teacher-approved synthetic task.

Write a question that can be answered by comparing a simple model against a **baseline**. Do not use private student records or real safety decisions.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines capstone data and **requirement**s.

Use your teacher-reviewed cumulative Plant Classifier records if they are complete and appropriate, or use the fully supplied synthetic parcel kit below as the bounded practice capstone. Choose and name one dataset/task version rather than merging their labels.

The kit instantiates the 72-record grid already described in the teacher guide; its clear/blocked categories are invented operational labels, not plant diagnosis or real parcel safety. Its numerical results differ from the separate illustrative 30/10 training-count example. Improvement is not required for a successful evidence-based project.

Baseline-first method and illustrative extension

Build the **baseline** before the candidate. For classification, use the most common **training** class, with a declared tie policy.

Choose one simple candidate using techniques already learned, such as k-nearest neighbours with **training**-only scaling and a fixed k. Record every development attempt; complexity earns no automatic credit.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines capstone models and **baseline**s.

▼ Earlier wording, alternatives and picture transcripts

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



MYSTERY The team must use capstone dataset and **baseline** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

Apply the connected idea: Capstone models and baselines

MAIN LESSON · SECTION 3

Read the actual kit counts before the illustrative examples

> Do not copy an example score as the result of a different dataset.

Complete actual kit role/count record

Actual supplied kit role	Rows / distinct parcel groups	Public label availability
Training	40 /40	22 clear, 18 blocked; fit the constant here
Validation	20 /20	9 clear, 11 blocked; score fixed predictions here
Reserved final	12 /12	Features supplied, label cells blank in student script; teacher key held separately
All source IDs	P00–P71; group_idequalsoriginal parcel ID	No source group crosses roles; IDs are audit metadata, not predictive inputs

Actual kit baseline chooses clear from 22 versus18 training labels. It gets 9/20=45%validation accuracy and finds 0/11 blocked cases. Precision is unavailable because it predicts no positives; do not call it perfect. This does not match the separate illustrative baseline 14/20=70%, whose training counts are 30/10. Keep every dataset/version and table labelled.

The kit is a hand-constructed synthetic grid, not a random or representative population sample. The full student program below includes every row; final reference cells are intentionally blank.

Original task proposal and dataset preparation context

Example proposal: can a small classifier improve validation accuracy over the **training**-majority **baseline** on fictional parcel measurements?

Before modelling, assign 40 **training**, 20 validation and 12 final-test records, with related parcel groups kept together.

Define blocked as positive, keep a fixed label rule, and reserve test labels with the teacher until Week 35.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Your teacher can prepare the 72-record fictional grid described in the teacher guide, or approve another documented synthetic dataset. Each supplied record has an ID, group ID, width and height in cm.

Training and validation labels are released for development; final-test labels stay with the teacher until the model is frozen.

The numerical results in later worked examples are illustrative and are not expected scores for your project.

▼ Optional separate illustrative30/10baseline/candidate result

Illustrative **training** has 30 clear and 10 blocked cases, so the **baseline** always predicts clear. On 20 validation cases with 14 clear and 6 blocked, its matrix is[[14, 0],[6, 0]], accuracy 70 percent.

Candidate matrix[[12, 2],[2, 4]] gives accuracy 80 percent, precision 4/6 and recall 4/6. These are example results, not the student team results.

```
# A check of the illustrative validation matrix.
# Replace with your own saved counts; do not invent results.
tn, fp, fn, tp = 12, 2, 2, 4
n = tn + fp + fn + tp
print("accuracy", (tn + tp) / n)
print("precision", tp / (tp + fp))
print("recall", tp / (tp + fn))
# Compare with the training-majority baseline: 14 / 20.
```

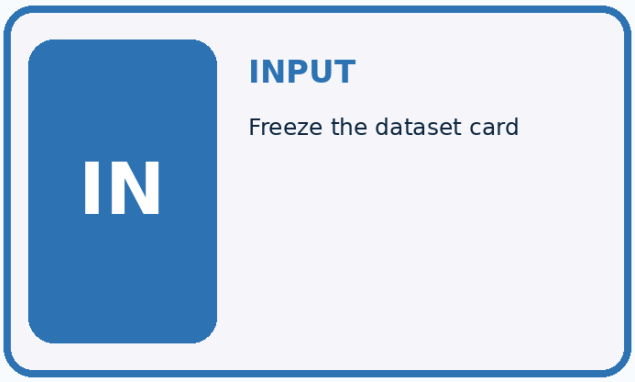
Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

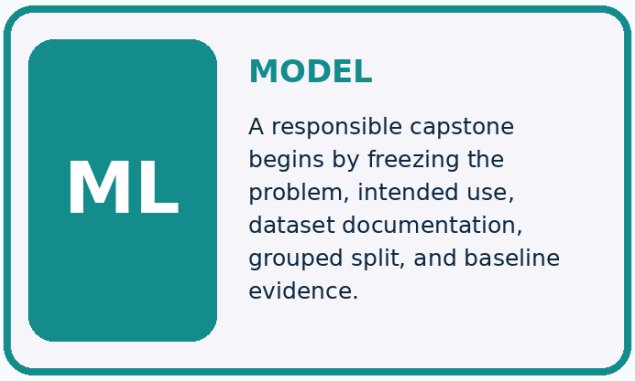
▼ Earlier wording, alternatives and picture transcripts

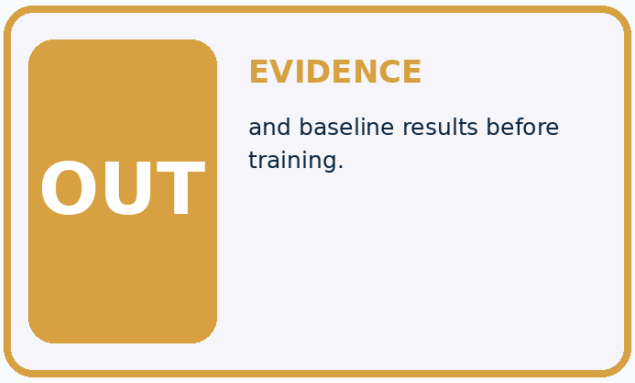
ML **MECHANISM**

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







QUESTION **Freeze** the dataset card, split, and **baseline result**s before **training**.

MODEL IDEA A responsible capstone begins by freezing the problem, intended use, dataset documentation, grouped split, and **baseline evidence**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD **Freeze** dataset card v3 and grouped split s2 with no plant identity crossing sets. Test set remains sealed; validation supports model choices.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Apply the connected idea: Capstone models and baselines

Connect the picture and the explanation

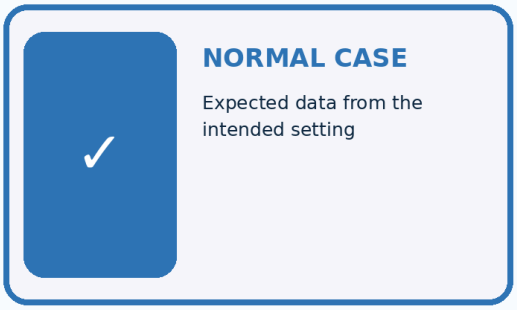
Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

MAIN LESSON · SECTION 4

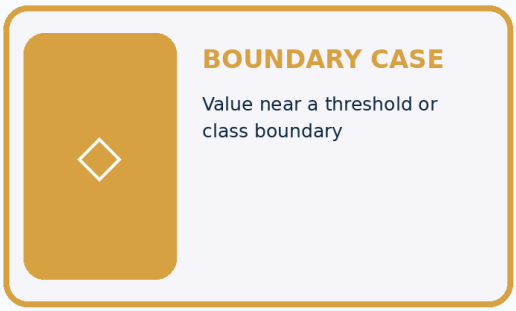
Compare a documented split with a misleading extra filename

> Source lineage follows a crop or derived view even after renaming.

Four evaluation conditions



NORMAL Expected data and expected behavior.

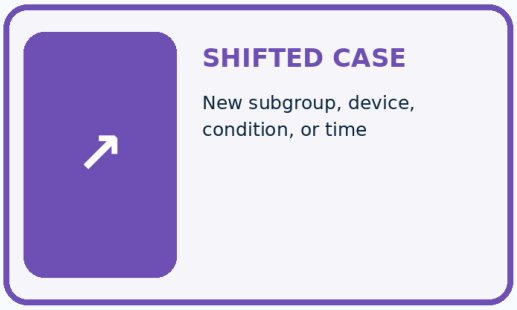


Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.



MISSING Necessary field, label, source, or measurement is unavailable.



SHIFTED New lighting, species, device, subgroup, or time condition.

Keep the claim bounded

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Choose a revealing case

- Which case most threatens today's claim, and why?

Predict the supplied changed case

- Change: A derived crop of plantG is added to **training**.

▼ **Check after predicting**

Prediction: The final group separation is violated.

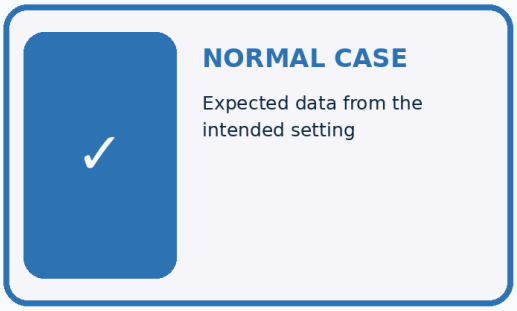
Reason: The crop retains the same source plant identity even with a new file name.

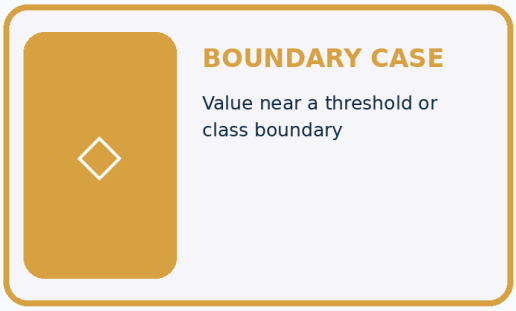
▼ Earlier wording, alternatives and picture transcripts

EVALUATION DESIGN

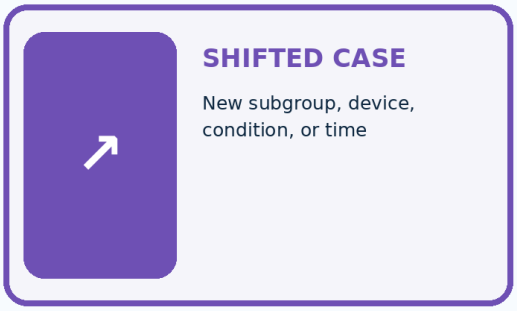
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Keep the original source group behind every derived view

> A correctly grouped small split still has sample-size limits.

Complete pictured source counts and separate-plan scope

Plan the plant capstone around independent groups and a **baseline** before collecting more rows.

Toy proposed dataset: 8 plants, 3 photos each. Development groupsA–F; final groupsG, H. Target is visible-spot status, with review for hidden leaves.

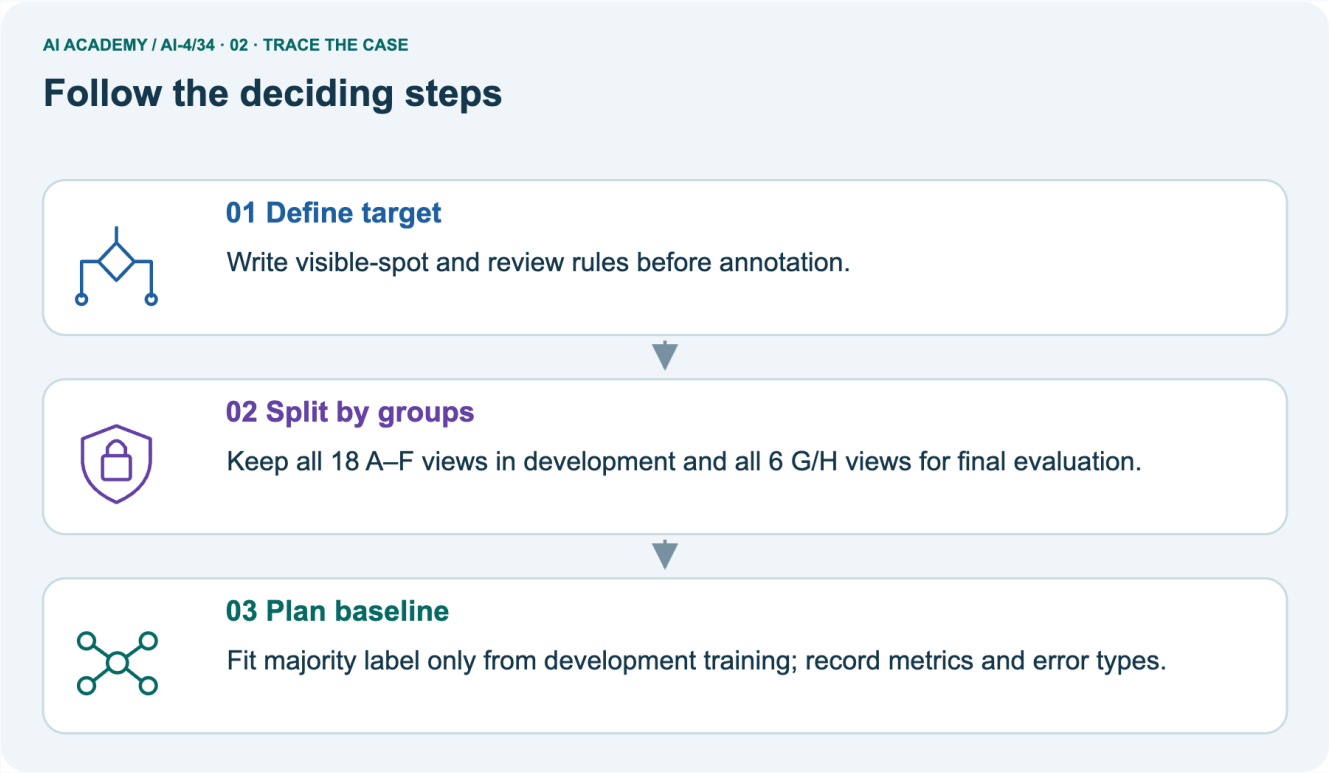
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate small plant plan:8 plants×3 views=24 photos; A–Fdevelopment18 views, G/Hfinal6 views. It is not the 72-parcel kit, whose groups are single-record source parcels. Within plant development, choose and record a further training/validation

assignment before fitting; the picture does not supply that assignment.

A derived crop ofGstill belongs to source plantGand cannot enter training whileGis held out. Two final plants provide little population evidence. The actual kit and the independent 40-related-pair task each have their own group/count contracts.

Trace target,groups and training-only baseline



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Define target	Write visible-spot and review rules before annotation.
2. Split by groups	Keep all 18 A–F views in development and all 6 G/H views for final evaluation.
3. Plan baseline	Fit majority label only from development training; record metrics and error types.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

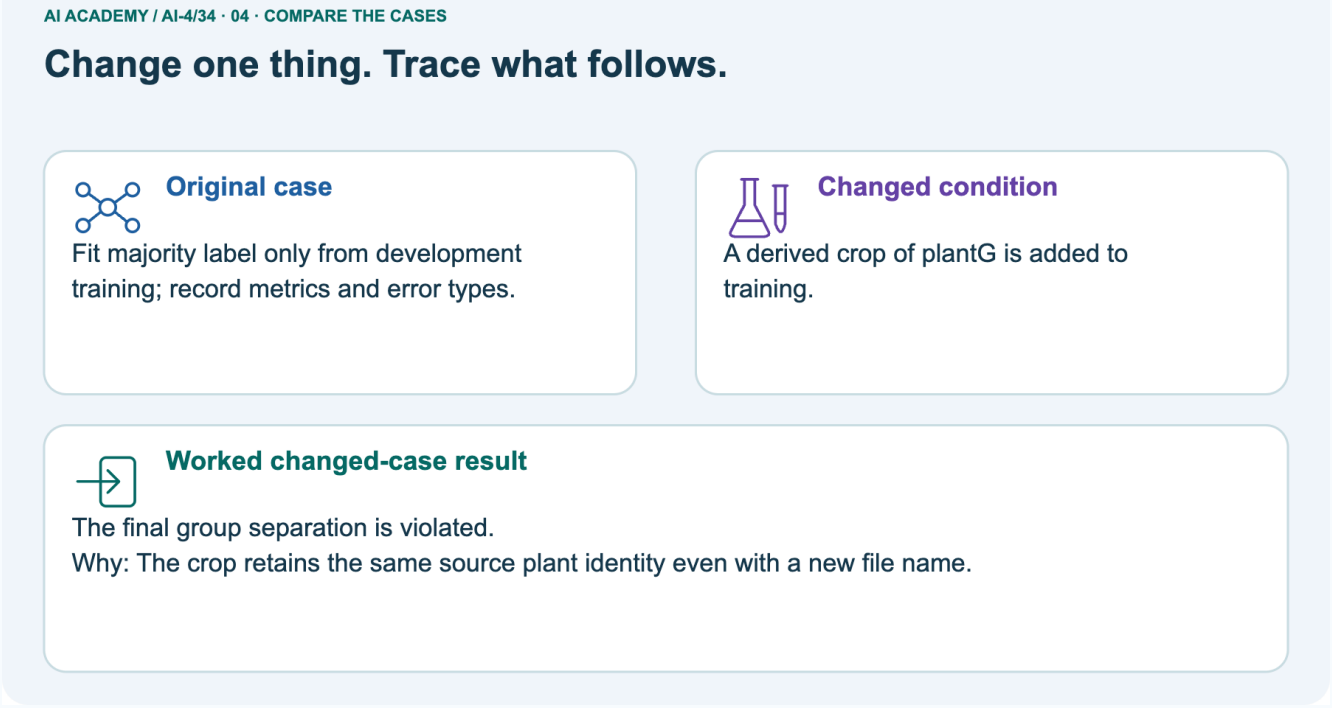
Read the picture in order. Fit majority label only from development **training**; record metrics and error types. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Predict a crop of held-outGentering training

- Change one thing: A derived crop of plantG is added to **training**. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain lineage and narrow-evidence limits

- A well-planned split does not create sufficient sample size; two held-out plants support only a very narrow claim.

- Read the labelled picture: What provenance field prevents the crop from appearing independent? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Use the corrected capstone timeline and ID roles

Capstone data and requirements at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Week 34	Task,data,split and baseline	Teacher checks feasibility
Week 35	Train,validate,analyze and freeze	Record candidate choices
End Week 35 / Week 36	Teacher-gated final check and defense	Evidence plus limitations

The data dictionary lists record ID, group ID, width in cm, height in cm and reference class. IDs track provenance and splitting; they are not automatic predictive inputs. Check uniqueness, units, missing values, class coverage and group separation. Fit imputation or scaling only on training.

Explain why record ID, group ID and measured width have different roles. Explain the steps or evidence behind your response.

▼ Optional separate illustrative candidate metrics

Capstone models and baselines at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Training	Baseline always clear	30 clear and 10 blocked
Validation baseline	14 correct of 20	6 blocked cases missed
Validation candidate	16 correct of 20	2 false alarms 2 misses

The candidate gains two correct cases while introducing two false alarms and reducing misses from six to two. Whether that tradeoff is acceptable depends on the predeclared classroom criteria. Do not choose a threshold or metric after the final test. If using a regression alternative, compare with a training-derived constant and report MAE in target units.

> **Compute the worked candidate accuracy, precision and recall. Explain the steps or evidence behind your response.**

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Four-step worked evidence for Week 34: Capstone Dataset and Baseline

STEP	EVIDENCE AND REASONING
1	Freeze dataset card v3 and grouped split s2 with no plant identity crossing sets.
2	Test set remains sealed; validation supports model choices.
3	Validation has 12 healthy and 8 spotted leaves. The always-healthy baseline scores 12/20=60% accuracy and 0/8=0% spotted recall. The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.
4	Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

STEP 1 **Freeze** dataset card v3 and grouped split s2 with no plant identity crossing sets.

STEP 2 Test set remains sealed; validation supports model choices.

STEP 3 Validation has 12 healthy and 8 spotted leaves. The always-healthy **baseline** scores 12/20=**60%** accuracy and 0/8=0% spotted recall. The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.

STEP 4 Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

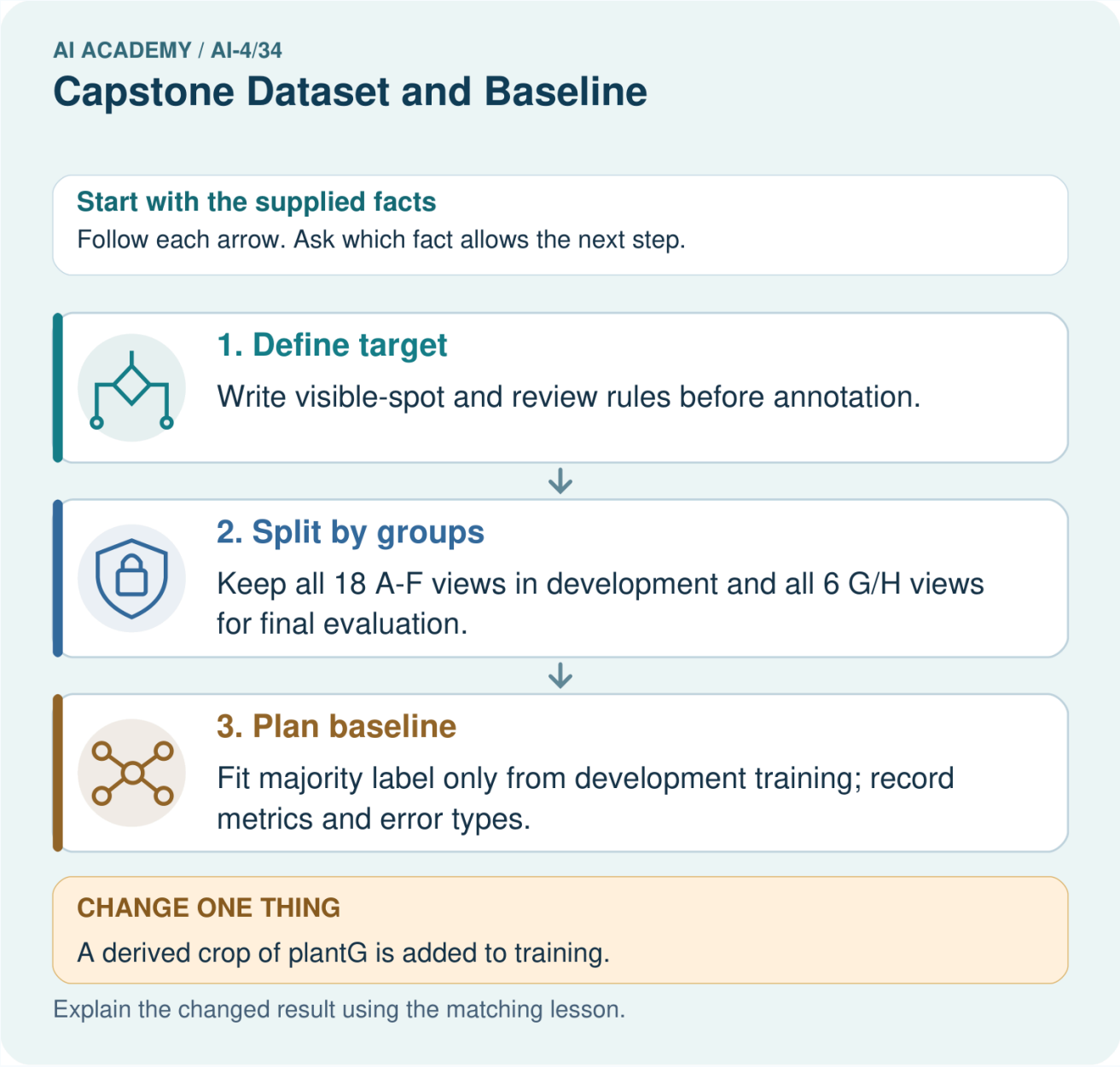
Apply the connected idea: Capstone models and baselines

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone Dataset and Baseline

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a project begun before its data and baseline are reviewable

> A baseline loss or pipeline failure is evidence to retain,not a reason to invent success.

Diagnose the main claim

- Diagnose one cause and name the cleanest next experiment:

CLAIM TO CHALLENGE Capstone work should begin by **training** the most advanced available model.

Explain capstone evidence and acceptance limits

- Submit a one-page proposal, **data dictionary**, split-membership record and risk/limitations note. The teacher approves feasibility and synthetic-data scope before modelling.

  If there are too few valid records for an independent test, use a clearly labelled demonstration and explicitly limit the claim rather than inventing **evidence**.

> **What should the team do if it cannot obtain a valid untouched final test?**

Baseline/candidate deliverable and freeze question

The deliverable is a runnable or fully traceable **baseline**/candidate comparison, a validation table and a signed-off **freeze** record. If the **baseline** wins, retain it and explain why.

If a pipeline fails, record the failure and fix it on development data before requesting final testing.

List what must be saved before the teacher releases final-test labels.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

</details>

A closer explanation

Apply the connected idea: Capstone models and baselines

MAIN LESSON · SECTION 8

Practise the data dictionary and baseline record with less help

> An ID identifies a record;a group ID tracks shared origin;a feature is a permitted measurement.

Try supported field-definition W2

Use workbook W2 for the connected example “Capstone data and **requirement**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Keep process criteria and evidence even if the model loses

Acceptance criteria describe the **process** even if the model loses: no split leakage; **prediction**s for every planned case or documented failures; the same metric for **baseline** and candidate; precision/recall with denominators; a model card with limits.

Improvement is a research outcome, not a grade **requirement**.

Save the data version, feature order, pre**process**ing values, model settings, tie policy and **prediction**s. A **prediction ledger** includes row ID, split, reference, **baseline output**, candidate **output** and error type.

Test labels remain held by the teacher. A peer should be able to reproduce one row by hand.

Explain the project record behind the result

- Use the worked case for Capstone data and **requirement**s. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain source provenance in the picture

- What provenance field prevents the crop from appearing independent? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
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Reserve the independent80-row/40-pair design

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Additional illustrative baseline practice**

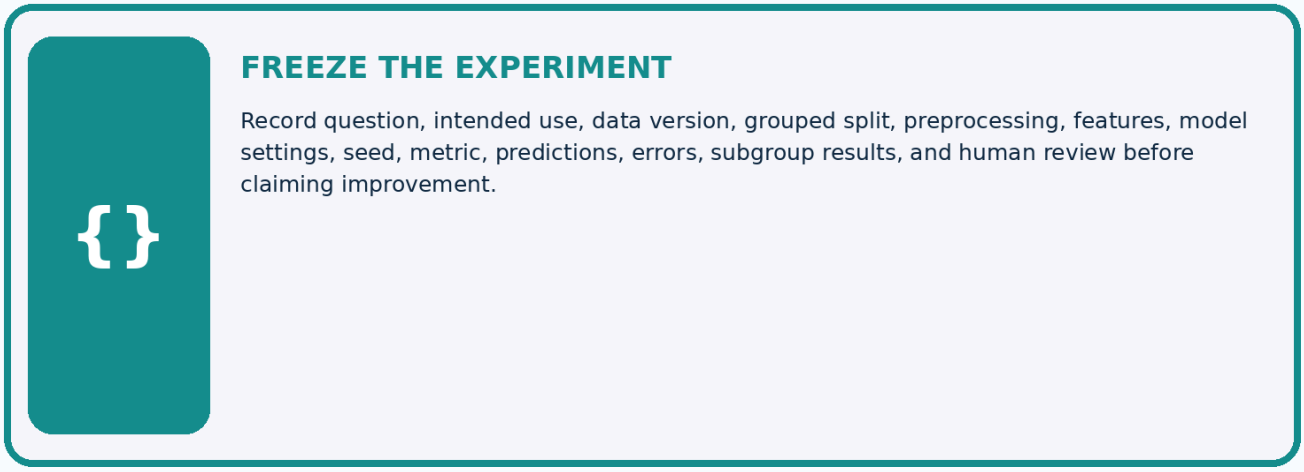
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A final answer is enough

▼ **Earlier wording, alternatives and picture transcripts**

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Apply the connected idea: Capstone models and baselines

Practise with less help: Define fields

Check your reasoning

Practise the connected topic: Capstone models and baselines

Explain the picture

MAIN LESSON · SECTION 9

Run the actual data-and-baseline gate with a complete kit

> The program includes all72student rows and never scores the blank final labels.

Complete data dictionary and baseline contract

Supplied field	Definition / valid use
id	P00–P71 unique row identity; use to align outputs, not as a numeric feature.
group_id	Original fictional parcel identity; all derived views must retain it. Here each supplied group has one row.
split	Saved training/validation/finalmembership; do not randomly resplit after seeing outcomes.
width_cm, height_cm	Nonnegative observed dimensions available before prediction, in cm; declare order, widththenheight.
label	Fictional clear/blocked reference for training/validation; blank for final in all student copies until the teacher-gated frozen-model step.
Baseline contract	Training majority, clear on a tie; accuracy and blocked recall with counts; no final scoring without released references.

Complete copyable,runnable student kit

```
import csv
from io import StringIO
from collections import Counter

# Complete student kit; final reference cells are deliberately blank.
DATA = """id,group_id,split,width_cm,height_cm,label
P00,P00,training,2,2,clear
P01,P01,training,3,2,clear
P02,P02,validation,4,2,clear
P03,P03,final,5,2,
P04,P04,training,6,2,clear
```

```
P05,P05,training,7,2,clear
P06,P06,validation,8,2,clear
P07,P07,final,9,2,
P08,P08,training,10,2,blocked
P09,P09,training,2,3,clear
P10,P10,validation,3,3,clear
P11,P11,training,4,3,clear
P12,P12,training,5,3,clear
P13,P13,validation,6,3,clear
P14,P14,final,7,3,
P15,P15,training,8,3,clear
P16,P16,training,9,3,blocked
P17,P17,validation,10,3,blocked
P18,P18,training,2,4,clear
P19,P19,training,3,4,clear
P20,P20,validation,4,4,clear
P21,P21,final,5,4,
P22,P22,training,6,4,clear
P23,P23,training,7,4,clear
P24,P24,validation,8,4,blocked
P25,P25,final,9,4,
P26,P26,training,10,4,blocked
P27,P27,training,2,5,clear
P28,P28,validation,3,5,clear
P29,P29,training,4,5,clear
P30,P30,training,5,5,clear
P31,P31,validation,6,5,clear
P32,P32,final,7,5,
P33,P33,training,8,5,blocked
P34,P34,training,9,5,blocked
P35,P35,validation,10,5,blocked
P36,P36,training,2,6,clear
P37,P37,training,3,6,clear
P38,P38,validation,4,6,clear
P39,P39,final,5,6,
P40,P40,training,6,6,blocked
P41,P41,training,7,6,blocked
P42,P42,validation,8,6,blocked
P43,P43,final,9,6,
P44,P44,training,10,6,blocked
P45,P45,training,2,7,clear
P46,P46,validation,3,7,clear
P47,P47,training,4,7,clear
P48,P48,training,5,7,blocked
P49,P49,validation,6,7,blocked
P50,P50,final,7,7,
P51,P51,training,8,7,blocked
P52,P52,training,9,7,blocked
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P53,P53,validation,10,7,blocked
P54,P54,training,2,8,clear
P55,P55,training,3,8,clear
P56,P56,validation,4,8,blocked
P57,P57,final,5,8,
P58,P58,training,6,8,blocked
P59,P59,training,7,8,blocked
P60,P60,validation,8,8,blocked
P61,P61,final,9,8,
P62,P62,training,10,8,blocked
P63,P63,training,2,9,clear
P64,P64,validation,3,9,blocked
P65,P65,training,4,9,blocked
P66,P66,training,5,9,blocked
P67,P67,validation,6,9,blocked
P68,P68,final,7,9,
P69,P69,training,8,9,blocked
P70,P70,training,9,9,blocked
P71,P71,validation,10,9,blocked
"""
records = list(csv.DictReader(StringIO(DATA)))
for row in records:
    row['width_cm'] = int(row['width_cm'])
    row['height_cm'] = int(row['height_cm'])
train = [r for r in records if r['split'] == 'training']
valid = [r for r in records if r['split'] == 'validation']
final_features = [r for r in records if r['split'] == 'final']
assert len({r['id'] for r in records}) == len(records) == 72
sets = [{r['group_id'] for r in part} for part in [train, valid, final_features]]
assert not (sets[0] & sets[1] or sets[0] & sets[2] or sets[1] & sets[2])
assert all(not r['label'] for r in final_features)
counts = Counter(r['label'] for r in train)
constant = 'blocked' if counts['blocked'] > counts['clear'] else 'clear'
predictions = [constant] * len(valid)
correct = sum(p == r['label'] for p, r in zip(predictions, valid))
positives = sum(r['label'] == 'blocked' for r in valid)
found = sum(p == r['label'] == 'blocked' for p, r in zip(predictions, valid))
print('roles', len(train), len(valid), len(final_features))
print('training clear/blocked', counts['clear'], counts['blocked'])
print('frozen baseline', constant)
print('validation correct/total', correct, len(valid))
print('validation blocked found/total', found, positives)
print('final references absent: not scored')
# IDs/groups are audit metadata, not model features. Tie rule chooses clear.
```

Concrete Gate1evidence checks

Gate1 evidence to audit	Concrete check with the supplied kit
Task and source	State toy method comparison, synthetic grid and excluded real uses. Keep dataset version and owner/reviewer fields.
Source integrity	72 unique IDs, role counts 40/20/12, no group overlap. A new crop inherits source group, not a new independent group.
Prediction inputs	Width/height in declared order and units; exclude IDs, split names and labels from candidate input features.
Baseline	Fit clear from 22/18 training counts; validation 9/20 correct, 0/11 blocked found. Retain all 20 predictions.
Final-label boundary	Student script contains no final references. Save feature/prediction membership and freeze choices before teacher release at the later gate.
Status and repair	Record actual check outcomes and unresolved issues; do not invent a teacher sign-off. An honest losing baseline/candidate is an acceptable research result.

Audit,run or trace,and repair

1. Capstone Gate1: run or fully trace the complete supplied kit/baseline program, then exchange the data dictionary and evidence checklist with a peer. Inspect every failed assertion or mismatch before candidate modelling. Save the actual dataset choice/version, roles, labels, baseline predictions and validation counts.

   The kit’s teacher key and label-construction rule remain in teacher materials; the generated source is synthetic and a narrow protocol exercise. Use actual teacher review status, not a prefilled approval. Keep the independent related-pair plan for an uncoached first attempt.

Normal,tie,missing-final and derived-source probes

Complete gate probe	Expected bounded check
Normal supplied 72 rows	40 training, 20 validation, 12 final; baselineclear, validation 9/20.

Complete gate probe	Expected bounded check
Boundary a tie in a different training dataset	Use the predeclared clear-on-tie rule; do not consult validation to choose a constant.
Missing final references	Predict from features if desired, but report final score not evaluated. Do not use blanks as clear labels.
Derived crop of a held-out source	Keep parent group ID and role; new filename does not authorize a new split assignment.

Keep the evidence ledger

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Missing
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

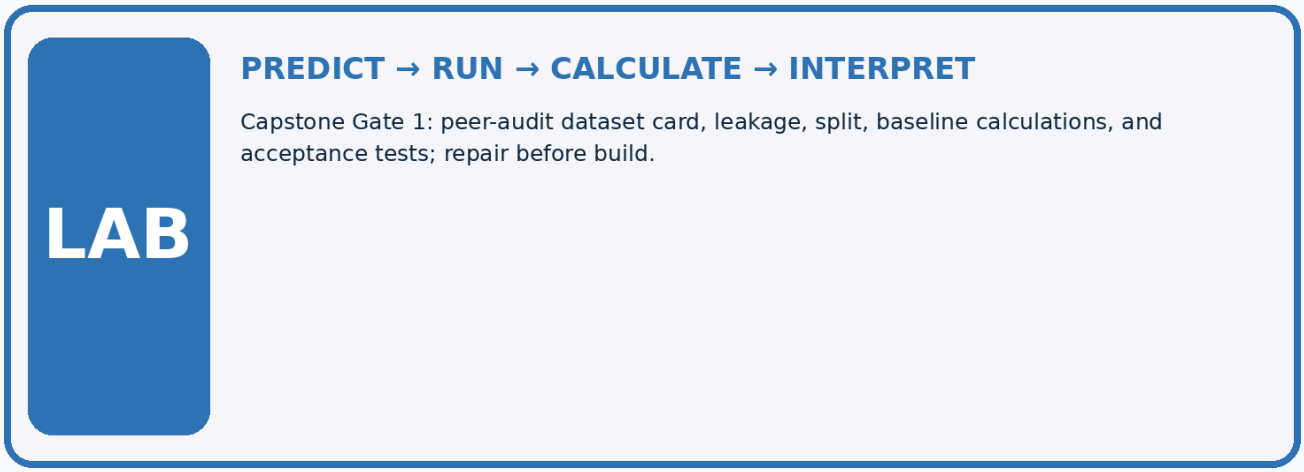
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

<details>
<summary>▼ Earlier wording, alternatives and picture transcripts</summary>

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Capstone Gate 1: peer-audit dataset card, leakage, split, **baseline** calculations, and **acceptance test**s; repair before build.

3. Run one normal, one boundary, one missing, and one shifted case.
</details>

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: repair a new related-pair data plan

> Use source identity and feature legitimacy before calculating a score.

Complete independent scenario and scope

A proposed 80-row dataset has 40 related pairs, and the team splits individual rows randomly then uses pair ID as a feature. Revise its plan.

The selected 80-row/40-related-pair task differs from the kit's72 single-record parcels. In this task both rows in each pair share one source group, so assign whole pairs to roles and exclude pair ID from predictive inputs. An exact 24/8/8 pair allocation is one possible declared example:48/16/16 rows, provided coverage and intended-use requirements are checked.

No class counts or baseline score are supplied; do not invent them.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply independently

Week 34 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Your capstone dataset includes duplicate photographs in both **training** and validation. What should be repaired before measuring the **baseline**?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Publish the frozen Plant Classifier design-and-data bundle with version IDs and approval record.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A capstone has two views of each of 40 leaves. Specify a 24/8/8 leaf split and **baseline** procedure. When may final labels be consulted?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Keep the design-and-data bundle stable for the candidate week

> The next experiment should cite this exact source and baseline record.

Project artifact and evidence

THIS WEEK'S CHECKPOINT Publish the frozen Plant Classifier design-and-data bundle with version IDs and approval record.

Retain one chosen capstone task/data version, a data dictionary/grouped split, dataset card, baseline run and full validation ledger. A reviewer records what was actually checked and unresolved issues. The supplied kit is complete for a reproducible toy exercise; it does not provide population representativeness or real plant/parcel suitability.

Carry the same saved student dataset into Week35 rather than regenerating labels or reshuffling evaluation rows.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Limits,uncertainty and human decision

Which **requirement** or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



Evidence for your project

MAIN LESSON · SECTION 16

Finish with a runnable baseline and an honest gate status

> Explain the chosen source,group roles and the final-label boundary.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Define a feasible capstone question and acceptance criteria

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the current experiment**

REFERENCE · SECTION 5

Reference · Words for the current experiment

> Tie each word to a particular input, fitted choice or evidence record.

Core vocabulary

Term	Precise meaning	My ML example
freeze	lock a reviewed version for comparison	
acceptance test	observable required behavior	
baseline result	simple method's measured performance	
evidence bundle	linked artifacts supporting a claim	

Closer-explanation terms

Word	Meaning
Requirement	A concrete condition the project must meet
Acceptance criterion	Evidence that decides whether a requirement is met
Data dictionary	Definitions and units for dataset fields

Word	Meaning
Milestone	A review point with a stated deliverable

▼ Extension vocabulary

Term	Meaning
Baseline	A simple reference method fitted or chosen from training
Prediction ledger	A per-record table of reference and predicted outputs
Frozen artifact	A saved model and procedure that will not change during testing
Decision record	A written reason for a development choice

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









MISCONCEPTION TO REPAIR Capstone work should begin by **training** the most advanced available model.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Terms for Capstone models and baselines

▼ Extra challenge · Build a stronger explanation

REFERENCE · SECTION 10

Extra challenge · Build a stronger explanation

> Choose the next useful challenge after the main route.

Thinking ladder

- DEFINE Use all four terms precisely: **freeze**, **acceptance test**, **baseline result**, **evidence bundle**.
- CALCULATE Reproduce the worked **evidence** with units and denominator.
- TRACE Explain the data, code or decision path.
- DIAGNOSE Capstone work should begin by **training** the most advanced available model.
- TRANSFER Solve a fresh dataset or model case.
- IMPROVE Change one cause and rerun fixed **evidence**.
- DEFEND State intended use, metric, limitation, and human responsibility.

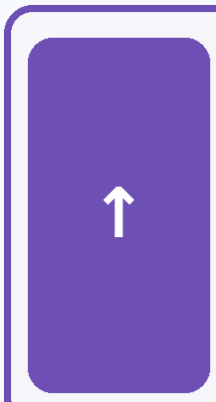
MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



ML BUILDER MASTERY

1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

▼ **Reference · Inspect your evidence**

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use after practice; keep the selected fresh80-row40-related-pair split repair for the individual attempt.

Experiment evidence checklist

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

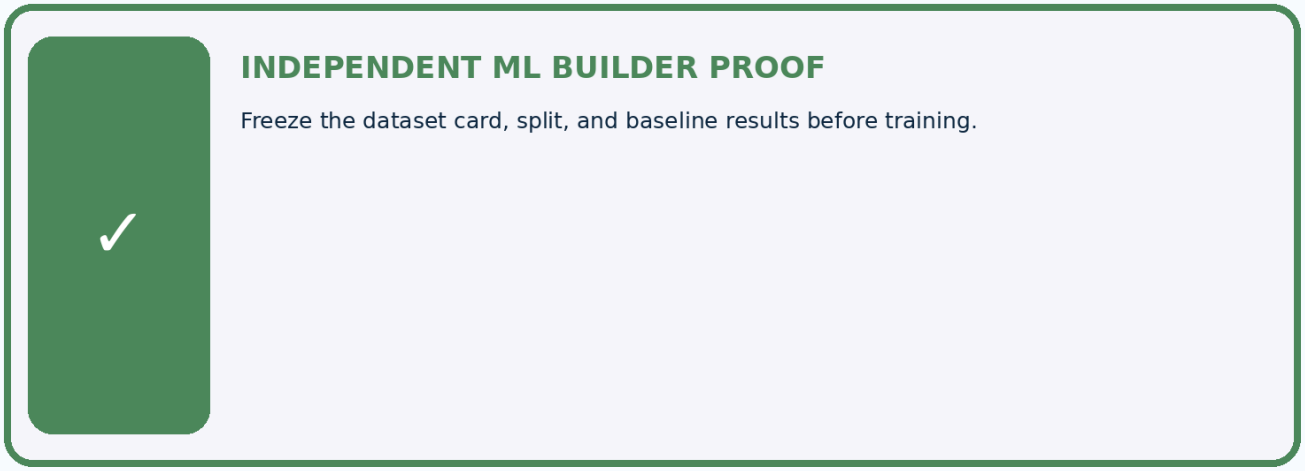
- Independent **evidence** and defense:
- My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



REQUIRED PROOF **Freeze** the dataset card, split, and **baseline result**s before **training**.

NEXT BRIDGE Carry the experiment **evidence** into: Capstone Train, Analyze, Improve.

▼ Extra practice · Earlier complete case

Extra practice · Earlier complete case

> Attempt the question before opening its worked explanation.

Situation and question

Capstone has60 independent leaves, several photos per leaf, healthy/spotted labels.

Plan grouping and **baseline**.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Keep all views together in train/validation/test partitions. Define labels/use, fit a majority **baseline** on **training** labels and evaluate declared metrics.

Evidence or stage	Value or result
Dataset	Sixty leaves, multiple views each
Group	Keep views of one leaf together
Partition	Separate train, validation and test
Baseline	Fit majority choice on training labels

▼ Earlier wording, alternatives and picture transcripts

Week 34 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Begin the capstone with independent groups

Capstone has60 independent leaves, several photos per leaf, healthy/spotted labels.

Plan grouping and **baseline**.

Keep all views together in train/validation/test partitions. Define labels/use, fit a majority **baseline** on **training** labels and evaluate declared metrics.

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Use one main route; revisit the checklist only when useful.

Checklist and first prediction

By the end, I can	How I will show it
Define a feasible capstone question and acceptance criteria	Explain the deciding step, show a trace or test, and state one limit.
Create a data and split plan before modelling	Explain the deciding step, show a trace or test, and state one limit.
Specify evidence, privacy and a fallback	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Plan the plant capstone around independent groups and a **baseline** before collecting more rows.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Plan the plant capstone around independent groups and a **baseline** before collecting more rows.
- Which labelled step matches this detail? Fit majority label only from development **training**; record metrics and error types.
- 1. Define target
- 2. Split by groups
- 3. Plan **baseline**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Plan **baseline** Fit majority label only from development **training**; record metrics and error types. Read the picture in order.

Fit majority label only from development **training**; record metrics and error types. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: A derived crop of plantG is added to **training**. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. The final group separation is violated.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The final group separation is violated. The crop retains the same source plant identity even with a new file name.

Repair and keep the record

Review Model Cards and Human Review (ai-4/33). Goal: Write a model card whose claims match **evidence**. Short practice: Write a bounded replacement for the publicity sentence.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Subgroup Performance (ai-4/31). Goal: Calculate subgroup results with sample sizes. Short practice: Why should the report retain subgroup rates alongside the total?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learning Curves (ai-4/28). Goal: Read **training**-size learning curves. Short practice: What comparison is valid if the validation set changes between sizes?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

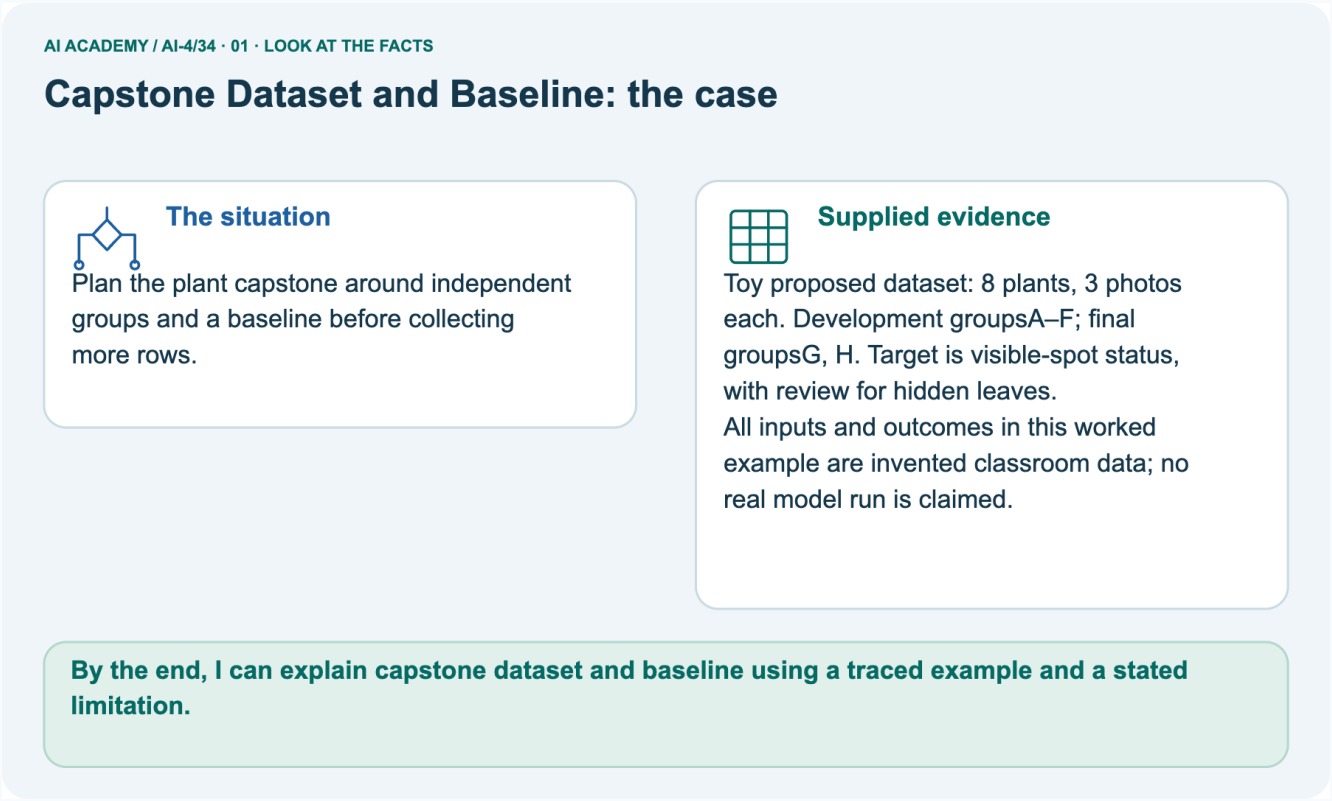
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the current question.

01. Capstone Dataset and Baseline: the case



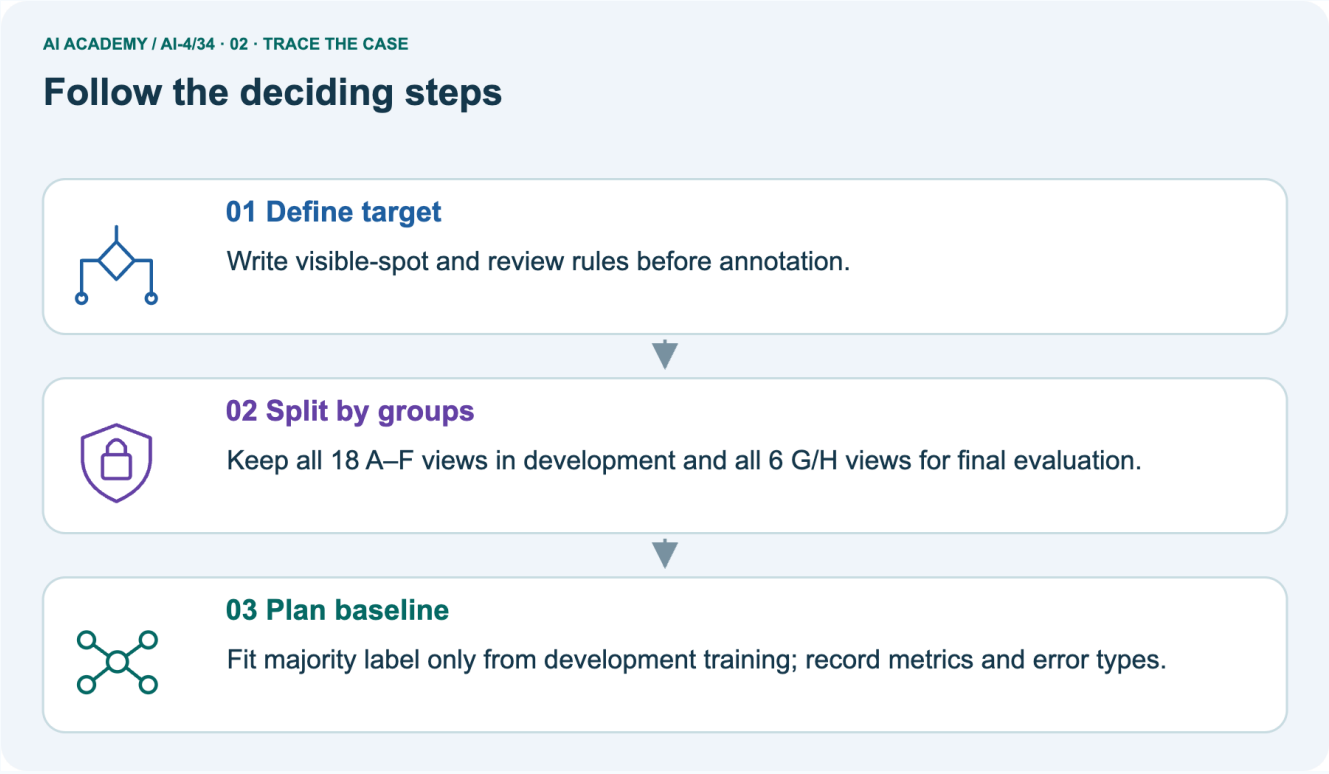
Follow the labelled picture with the matching explanation. Open the vector diagram

Plan the plant capstone around independent groups and a **baseline** before collecting more rows. Supplied facts: Toy proposed dataset: 8 plants, 3 photos each. Development groupsA–F; final groupsG, H.

Target is visible-spot status, with review for hidden leaves. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Define target: Write visible-spot and review rules before annotation. → Split by groups: Keep all 18 A–F views in development and all 6 G/H views for final evaluation. → Plan **baseline**: Fit majority label only from development **training**; record metrics and error types.

Point to each numbered stage and explain which recorded **input** it uses.

03. Four-step worked evidence for Week 34: Capstone Dataset and

Baseline

AI ACADEMY / AI-4/34 · 03 · INSPECT THE LESSON MODEL

Four-step worked evidence for Week 34: Capstone Dataset and Baseline

Step	Evidence and reasoning
1	Freeze dataset card v3 and grouped split s2 with no plant identity crossing sets.
2	Test set remains sealed; validation supports model choices.
3	Validation has 12 healthy and 8 spotted leaves. The always-healthy baseline scores 12/20=60% accuracy and 0/8=0% spotted recall. The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.
4	Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

Follow the labelled picture with the matching explanation. Open the vector diagram

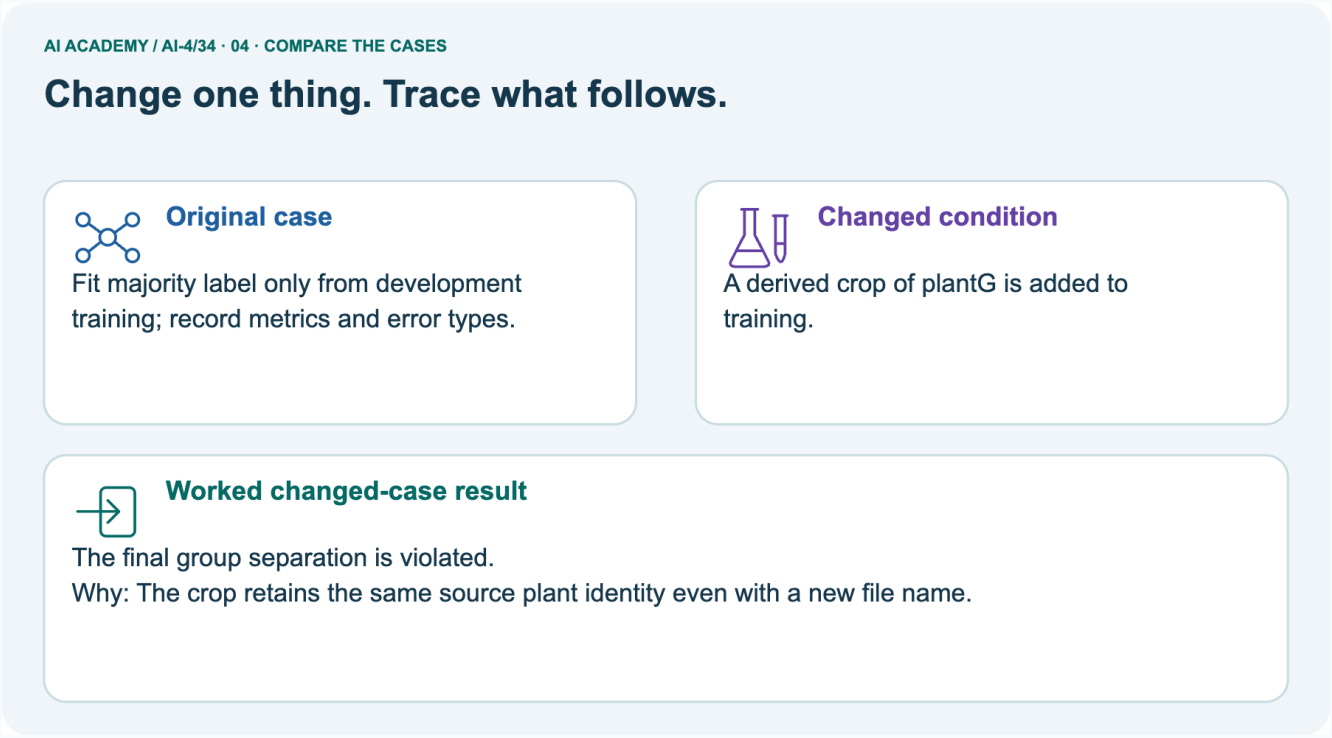
Four-step worked **evidence** for Week 34: Capstone Dataset and **Baseline**. 1 | **Freeze** dataset card v3 and grouped split s2 with no plant identity crossing sets.. 2 | Test set remains sealed; validation supports model choices.. 3 | Validation has 12 healthy and 8 spotted leaves.

The always-healthy **baseline** scores 12/20=**60%** accuracy and 0/8=0% spotted recall.

The rule has TP=5, FN=3, TN=10, FP=2: accuracy 75% and spotted recall 62.5%.. 4 | Acceptance requires valid citations, metric calculations, subgroup review, and safe escalation.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



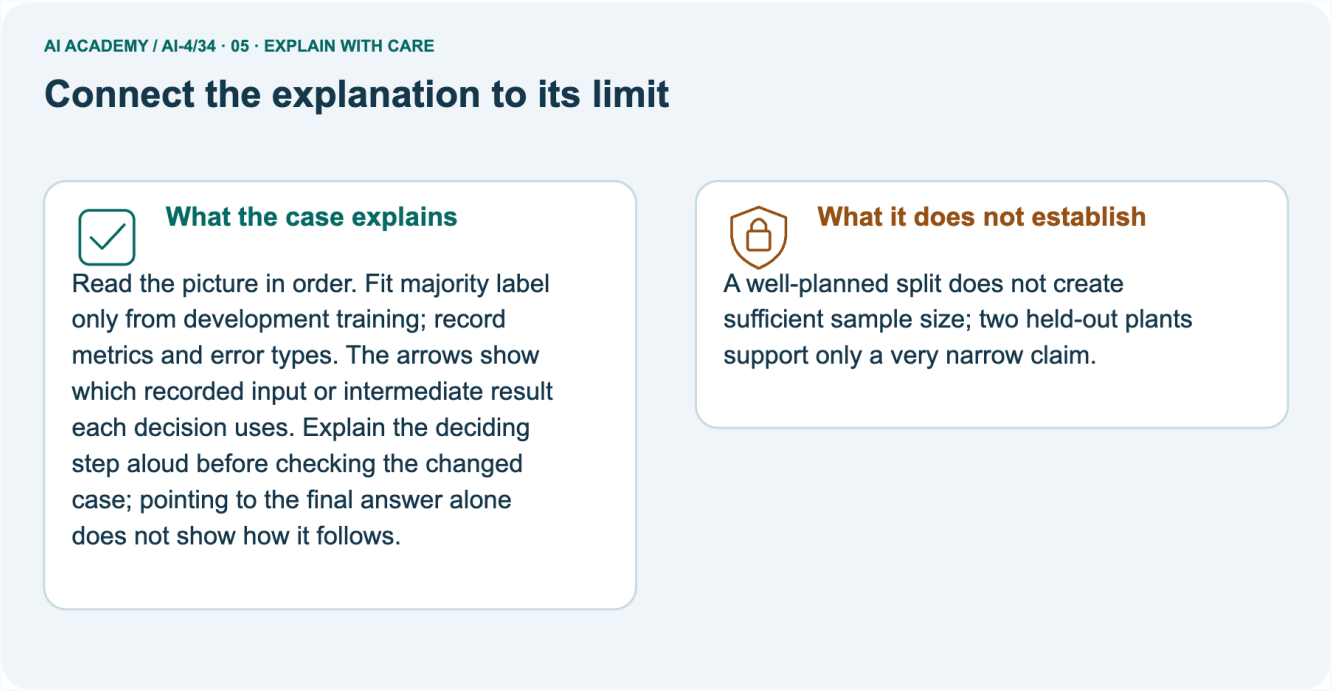
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Fit majority label only from development **training**; record metrics and error types. Changed condition: A derived crop of plantG is added to **training**. Expected result: The final group separation is violated.

Reason: The crop retains the same source plant identity even with a new file name.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

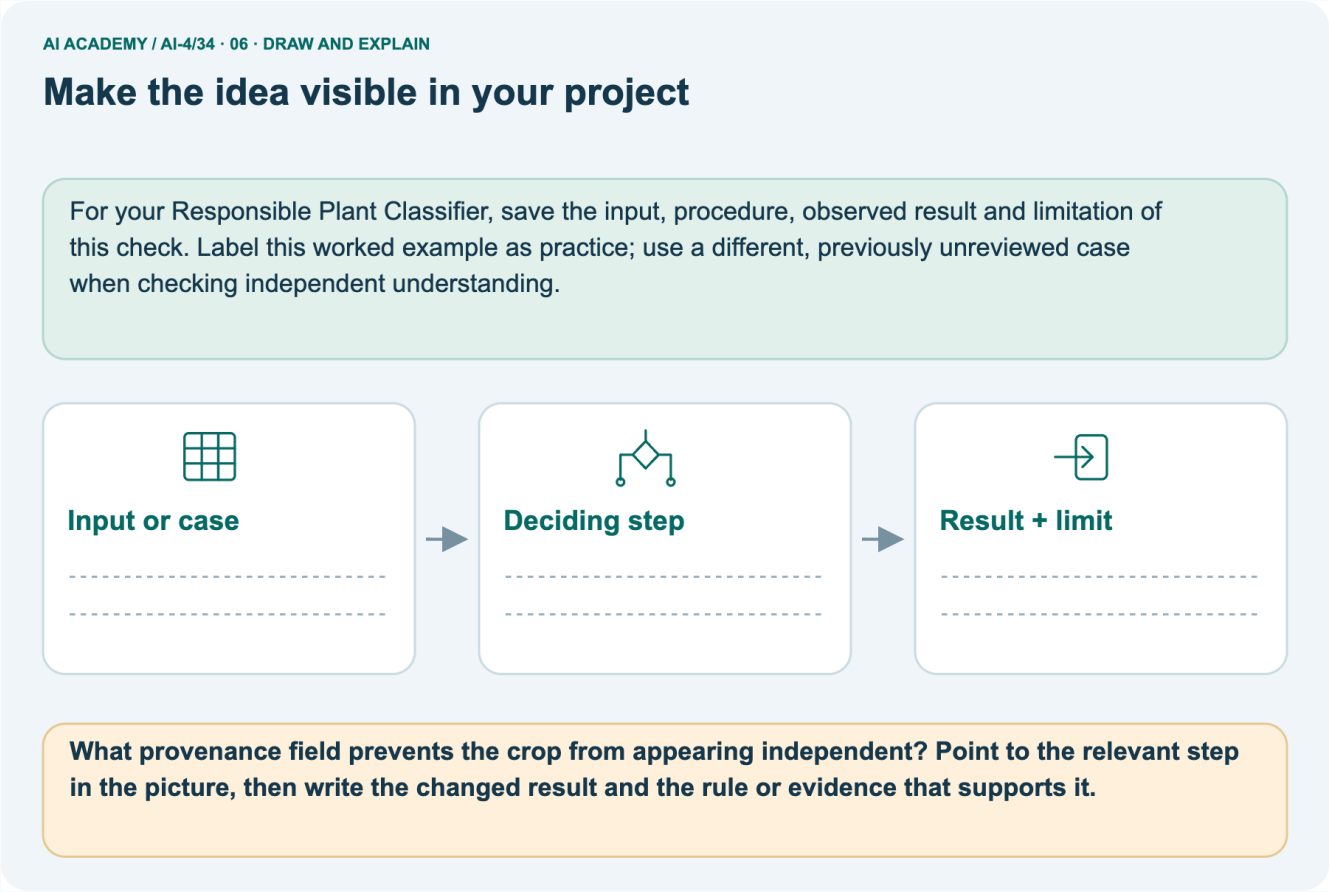
Read the picture in order. Fit majority label only from development **training**; record metrics and error types. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A well-planned split does not create sufficient sample size; two held-out plants support only a very narrow claim.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What provenance field prevents the crop from appearing independent? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Capstone Dataset and Baseline: the case

02. Follow the deciding steps

03. Four-step worked evidence for Week 34: Capstone Dataset and Baseline

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.